γ_5 in the Drell–Yan computation

When is $aa = vv$? Larin's scheme, open lines and pure singlet

DY_P2B — notes_Larin/

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Abstract

The vector boson $V = \gamma^*, Z$ couples to quarks through $\gamma^\mu(v_q + a_q\gamma_5)$. Squaring gives vector–vector, axial–axial and mixed pieces. We explain when the axial–axial piece equals the vector–vector one ($aa = vv$) and when it does not, what Larin's prescription for γ_5 does in squared amplitudes, and which axial-current renormalisation constants are needed. The rule: $aa = vv$ when V sits on the *same open quark line* in the amplitude and in the complex-conjugate amplitude, and only then. How to assemble all contributions with their counterterms: `assembly.pdf`.

Summary

Labels of the V vertex (as in `assembly.pdf`): v, a on open quark line 1 (flavour q); v_2, a_2 on open quark line 2 (flavour Q); $v_{\rm PS}$, $a_{\rm PS}$ on a closed quark loop, the **pure singlet (PS)**, with couplings $\sum_f v_f$, $\sum_f a_f$.

V in amplitude / conjugate	example	axial–axial
same open line	$v\,v^*$, $v_2 v_2^*$	$aa = vv$ (anticommuting γ_5)
different open lines	$v\,v_2^*$	$aa_2 \neq vv_2$, also in $d = 4$: Larin
identical quarks ($q = Q$)	both kinds, same couplings	split by line
closed loop (PS)	$v_{\rm PS}v^*$, $a_{\rm PS}a^*$	vector: compute; axial: Larin, $\propto \sum_f a_f$

The parity-odd pieces (va, av) contain one γ_5 in the hadronic trace and always need Larin (section 1).

1 The leptonic tensor

For massless leptons, $V \to \ell^-(l_1)\ell^+(l_2)$ with vertex $\gamma^\mu(v_\ell + a_\ell\gamma_5)$:

$$L^{\mu\nu} = \mathrm{Tr}\big[\not{l}_1\gamma^\mu(v_\ell + a_\ell\gamma_5)\not{l}_2\gamma^\nu(v_\ell + a_\ell\gamma_5)\big] = (v_\ell^2 + a_\ell^2)\,4\big(l_1^\mu l_2^\nu + l_1^\nu l_2^\mu - g^{\mu\nu} l_1\!\cdot\! l_2\big) + 2v_\ell a_\ell\,\mathrm{Tr}[\not{l}_1\gamma^\mu\not{l}_2\gamma^\nu\gamma_5]. \tag{1}$$

(The conjugate vertex is again $\gamma^\nu(v_\ell + a_\ell\gamma_5)$, since $\gamma^0(\gamma^\nu\gamma_5)^\dagger\gamma^0 = \gamma^\nu\gamma_5$; the first $(v_\ell + a_\ell\gamma_5)$ moves through $\not{l}_2\gamma^\nu$ without sign, and $(v_\ell + a_\ell\gamma_5)^2 = v_\ell^2 + a_\ell^2 + 2v_\ell a_\ell\gamma_5$.) The symmetric part multiplies the parity-even hadronic pieces (vv, aa, …), the antisymmetric (ϵ) part, with coupling $2v_\ell a_\ell$, the parity-odd ones (va, av, …).

Averaged over the lepton angles at fixed $q = l_1 + l_2$: $\langle l_1^\rho l_2^\sigma\rangle = A\,q^\rho q^\sigma + B\,q^2 g^{\rho\sigma}$; contracting with $g_{\rho\sigma}$ and $q_\rho q_\sigma$ gives $q^2/2 = (A + 4B)q^2$, $q^4/4 = (A + B)q^4$, so $A = \frac{1}{6}$, $B = \frac{1}{12}$: the antisymmetric part averages to zero and $\langle L^{\mu\nu}\rangle = \frac{4}{3}(v_\ell^2 + a_\ell^2)(q^\mu q^\nu - q^2 g^{\mu\nu}) = \frac{4}{3}(v_\ell^2 + a_\ell^2)L_1^{\mu\nu}$. The parity-odd pieces contribute only to observables that resolve the lepton direction; there they are projected with the antisymmetric projector L_6.

2 γ_5 in d dimensions; Larin's scheme

In $d \neq 4$ one cannot have both $\{\gamma_5, \gamma^\mu\} = 0$ and $\mathrm{Tr}[\gamma_5\gamma^\alpha\gamma^\beta\gamma^\gamma\gamma^\delta] \propto \epsilon^{\alpha\beta\gamma\delta} \neq 0$ (anticommutation and cyclicity force that trace to vanish). Anticommuting γ_5 is therefore unambiguous only if no trace with an odd number of γ_5 has to be computed.

Larin [hep-ph/9302240] replaces the axial vertex by

$$\gamma^\mu\gamma_5 \;\to\; \tfrac{1}{2}(\gamma^\mu\gamma_5 - \gamma_5\gamma^\mu) = \tfrac{i}{6}\,\epsilon^{\mu\nu\rho\sigma}\gamma_\nu\gamma_\rho\gamma_\sigma \tag{2}$$

(normalisation of arXiv:2306.10170, eq. (4.3)) and contracts the ϵ tensors in d dimensions. No γ_5 is ever moved, so the scheme is consistent; the axial current must then be renormalised by

$$Z_a^{ns} = 1 - 2C_F a + \Big(\tfrac{11}{2}C_F^2 - \tfrac{107}{36}C_F C_A + \tfrac{1}{18}C_F n_f + \frac{C_F\beta_0}{\epsilon}\Big)a^2, \tag{3}$$

$$Z_s = C_F T_R\Big(\frac{3}{2\epsilon} + \frac{3}{4}\Big)a^2 \quad \text{(singlet, per flavour)}, \tag{4}$$

with $a = \alpha_s/(2\pi)$, $\beta_0 = \frac{11}{6}C_A - \frac{2}{3}T_R n_f$. The finite parts restore the anticommuting-γ_5 Ward identities; the pole $C_F\beta_0/\epsilon$ is the $\overline{\mathrm{MS}}$ part. Conversion check (Larin's constants as I recall them, in $a_s = \alpha_s/4\pi = a/2$, to be checked against the original): $Z_5^{ns} = 1 - 4C_F a_s + a_s^2(22C_F^2 - \frac{107}{9}C_A C_F + \frac{2}{9}C_F n_f)$, $Z_{\overline{\mathrm{MS}}}^{ns} = 1 + a_s^2(\frac{22}{3}C_A C_F - \frac{4}{3}C_F n_f)/\epsilon$; with $a_s^2 = a^2/4$ these give the expression above. The singlet finite constant has $\frac{31}{18}C_F n_f$ instead of $\frac{2}{9}C_F n_f$: the difference $\frac{3}{2}C_F n_f a_s^2 = \frac{3}{4}C_F T_R n_f a^2$ is the $\frac{3}{4}$ of Z_s (arXiv:2306.10170, eqs. (5.5), (5.6)).

Remark on arXiv:2306.10170, eq. (5.13): the $\bar{G}^{(0),n}$ coefficient is printed as $2\beta_0/\epsilon$; expanding $Z_a^{ns}\sqrt{\bar{a}}$ gives $2C_F\beta_0/\epsilon$ ($C_F\beta_0/\epsilon$ from $(-2C_F)(-\beta_0/2\epsilon)$ and $C_F\beta_0/\epsilon$ from the pole of Z_a^{ns}): probably a missing C_F.

3 Same line: $aa = vv$

One open massless quark line, V attached once in the amplitude and once in the conjugate, any number of gluons. The spin-summed product is one trace,

$$\mathrm{Tr}\big[\not{p}\,\Gamma_1\,\gamma^\mu\gamma_5\,\Gamma_2\,\not{p}'\,\bar{\Gamma}_2'\,\gamma^\nu\gamma_5\,\bar{\Gamma}_1'\big], \tag{5}$$

Γ_2 = string from the V vertex to the end of the line in the amplitude, $\bar{\Gamma}_2'$ the same in the conjugate (reversed). Along a massless line propagators $\not{k}$ and vertices γ^α alternate and the line ends with a vertex, so $\Gamma_2 = \not{k}_1\gamma^{\alpha_1}\cdots\not{k}_j\gamma^{\alpha_j}$ has $2j$ matrices; the same for $\bar{\Gamma}_2'$. Between the two γ_5 there are $2j + 1 + 2j' + 1$ matrices: an *even* number. With anticommuting γ_5, moving one γ_5 next to the other costs $+1$, and $\gamma_5^2 = 1$:

$$\boxed{\mathrm{Tr}[\cdots\gamma^\mu\gamma_5\cdots\gamma^\nu\gamma_5\cdots] = \mathrm{Tr}[\cdots\gamma^\mu\cdots\gamma^\nu\cdots] \quad\Rightarrow\quad aa = vv} \tag{6}$$

diagram by diagram, in any d, also with loops on the line (gluon loops, quark loops without V). No trace with a single γ_5 is computed: the two are removed before tracing. Helicity version: $P_\pm\gamma^\alpha = \gamma^\alpha P_\mp$, $P_\pm\not{k} = \not{k}P_\mp$, so chirality h is conserved along the line, $A_a(h) = h\,A_v(h)$ and $\sum_h |A_a|^2 = \sum_h h^2|A_v|^2 = \sum_h |A_v|^2$. The parity-odd $\sum_h A_v A_a^* = \sum_h h|A_v|^2$ keeps one γ_5 in the trace: it needs Larin.

Larin breaks it at $O(\epsilon)$. Larin's vertex is not $\gamma^\mu\gamma_5$ in d dimensions. Born:

$$vv = N_c(d-2)z^{-3}, \qquad aa_{\mathrm{Larin}} = -\tfrac{1}{6}N_c(d-2)(d-3)(d^2-8d+10)z^{-3}, \tag{7}$$

$$\frac{aa_{\mathrm{Larin}}}{vv} = (1-2\epsilon)\Big(1 - \tfrac{2}{3}\epsilon^2\Big). \tag{8}$$

Equal at $d = 4$, different at $O(\epsilon)$; the same pattern holds for the double-real same-line pieces. Larin $+Z_a^{ns}$ and anticommuting γ_5 are both consistent and give the same physical non-singlet result; the $O(\epsilon)$ differences cancel only in the full combination (reals, virtuals, collinear counterterms in the same scheme).

Choice. Same-line parity-even pieces: anticommuting γ_5, i.e. compute vv and set $aa = vv$, at all orders (RRR, RR, R, B and the non-singlet part of the loop contributions). Then Z_a^{ns} is never needed for them. Everything else with an axial vertex (cross-line, PS, parity-odd): Larin $+Z$'s.

4 Different lines: $aa \neq vv$

$q\bar{q} \to VQ\bar{Q}$ with V on the q line in the amplitude and on the Q line in the conjugate: for $q \neq Q$ the product is two traces with one γ_5 each, and the two γ_5 cannot meet. With chiralities h_1, h_2 of the lines,

$$\sum_{h_1 h_2} A_a^{(1)} A_a^{(2)*} = \sum_{h_1 h_2} h_1 h_2 \, A_v^{(1)} A_v^{(2)*} \neq \sum_{h_1 h_2} A_v^{(1)} A_v^{(2)*} : \tag{9}$$

configurations with $h_1 = -h_2$ enter with opposite sign, so $aa_2 \neq vv_2$ already in four dimensions. Each trace has a single γ_5, so anticommuting γ_5 is ambiguous: these pieces need Larin (and Z_a^{ns} on both vertices in their loop corrections). Their coupling is $a_q a_Q$; summed over Q, $\sum_Q a_Q$ cancels within complete generations except for the b.

Identical quarks ($q\bar{q} \to Vq\bar{q}$): both lines have the same couplings, so v_q^2 and a_q^2 multiply same-line products, cross-line products and the exchange interference together. $aa = vv$ holds only for the same-line ones: the pieces must be separated by line, not by coupling.

5 Closed quark loops: pure singlet

V on a closed loop, summed over the n_f flavours of the loop: couplings $\sum_f v_f$, $\sum_f a_f$; only in loop amplitudes.

- **Vector PS.** Furry: zero with two gluons on the loop (V + 1 parton at one loop; four-quark processes at one loop; $q\bar{q} \to V$ at two loops). With three gluons on the loop (d^{abc}) it is non-zero: V + 1 parton at two loops, $q\bar{q} \to Vgg$ at one loop, $gg \to Vg$ at one loop. It does not cancel between flavours ($v_u \neq -v_d$).

- **Axial PS.** One γ_5 in the loop: the anomalous triangle; Larin $+Z_s$ is required. Example: the one-loop $a_{\rm PS}$ amplitude for $q\bar{q} \to Vg$ times the tree a^* (an NNLO RV piece) is non-zero by itself (Landau–Yang does not apply: one gluon is off shell); its coupling $a_q \sum_f a_f$ makes it vanish, together with its Z_s counterterm, in the sum over a complete mass-degenerate doublet ($a_u = -a_d$).

- With $n_f = 5$ massless flavours the b is left without its partner: $\sum_f a_f \neq 0$. A correct treatment of this remainder needs the massive top in the loop (arXiv:2306.10170, introduction); in DY it is usually small and treated separately. To check in the N^{3}LO DY literature before deciding.

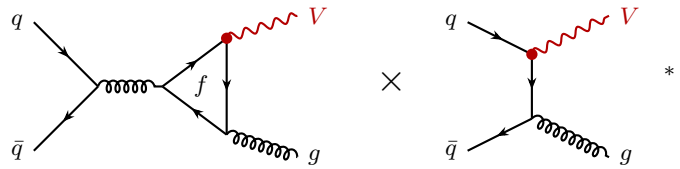
- $gg \to Vg$: vector $\propto d^{abc}$, axial $\propto f^{abc}$, so their interference (parity-odd) vanishes by colour.

- γ^*: $a_f = 0$. W: no PS (a W cannot couple to a loop that connects only to gluons).

6 Representative diagrams of the pieces with $aa \neq vv$

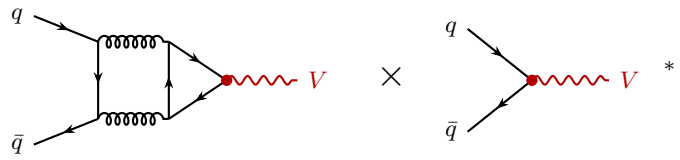
Each row shows the amplitude (left) times the complex-conjugate amplitude (right). The red dot marks the V vertex; the red wavy line is V, curly lines are gluons, arrows give the fermion flow. In all of them the two V vertices are *not* on the same open quark line, so the argument of section 3 does not apply.

(a) PS, one loop × tree: RV ($q\bar{q} \to Vg$), coupling $a_q \sum_f a_f$



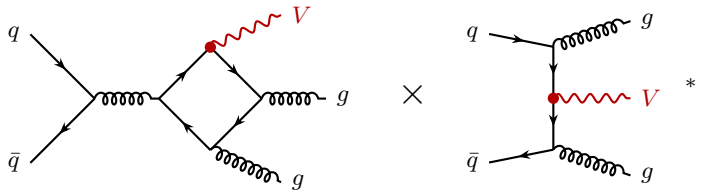
V on a closed loop of flavour f with two gluons: vector part $= 0$ (Furry), axial part = anomalous triangle.

(b) PS, two loops × tree: VV ($q\bar{q} \to V$), coupling $a_q \sum_f a_f$



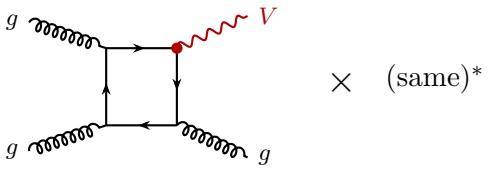
Again two gluons on the loop: vector $= 0$, axial $\neq 0$.

(c) PS with three gluons: RRV ($q\bar{q} \to Vgg$), couplings $v_q \sum_f v_f$ and $a_q \sum_f a_f$



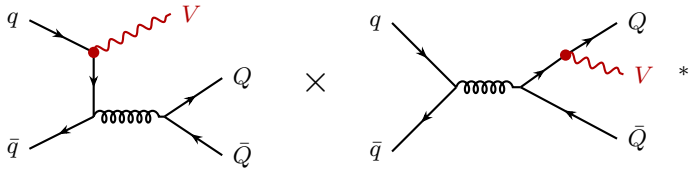
Three gluons on the loop: the vector part survives ($\propto d^{abc}$), the axial part $\propto f^{abc}$.

(d) Loop-induced: $gg \to Vg$, one loop × one loop, couplings $(\sum_f v_f)^2$, $(\sum_f a_f)^2$

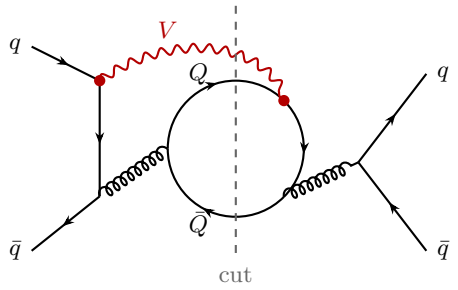


V on a closed loop on both sides; the vector–axial interference vanishes by colour ($d^{abc} f^{abc} = 0$).

(e) Cross-line, produced pair (PS-type): RR ($q\bar{q} \to VQ\bar{Q}$), couplings $v_q v_Q$, $a_q a_Q$

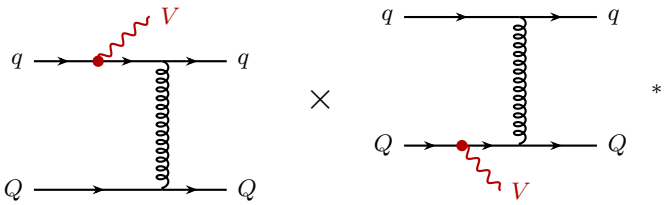


As a cut diagram (amplitude left of the cut, conjugate right of it) the Q line closes into a loop that contains V once and two gluons: the cut of the two-loop PS diagram (b).



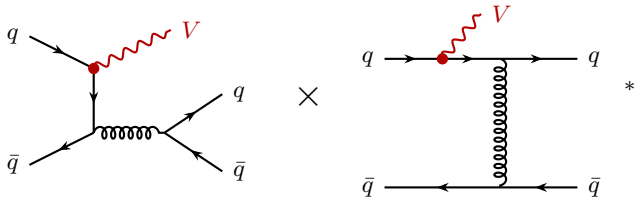
Three vector vertices on the closed Q loop: vv_2 integrates to zero (Furry; it is antisymmetric under $Q \leftrightarrow \bar{Q}$), aa_2 survives with coupling $a_q a_Q$, i.e. $a_q \sum_Q a_Q$ after the flavour sum: the same bucket as the PS.

(f) Cross-line, Q in the initial state: RR ($qQ \to VqQ$), couplings $v_q v_Q$, $a_q a_Q$



Two traces with one γ_5 each; no closed loop and no $Q \leftrightarrow \bar{Q}$ symmetry: $aa_2 \neq vv_2$, both non-zero in general. Own bucket (a_Q fixed by the parton distribution, not summed).

(g) Identical quarks, direct × exchange: RR ($q\bar{q} \to Vq\bar{q}$), coupling v_q^2, a_q^2



Annihilation-type amplitude times scattering-type amplitude: the product is one single trace through both lines, and the two V vertices are on different lines of the diagrams. With the same couplings (v_q^2, a_q^2) for every product, these terms are mixed with the same-line ones unless the pieces are separated by line.